

Online Retail Dataset – Summary Report

During data cleaning, several issues were identified and addressed:

- The CustomerID column contained 135,080 missing values. These records were retained because they represent anonymous customers and removing them would significantly reduce the dataset.
- The Description column had 1,454 missing values, which were replaced with "Unknown Product" to maintain data completeness.
- A total of 5,268 duplicate records were identified and removed.
- Date values were converted to a standard datetime format for time-series analysis.
- Text fields such as Description and Country were standardized using title case to ensure consistency.
- Records with negative quantities and prices were identified as returns, cancellations, or data-entry errors and were handled appropriately.

Key EDA Findings

1. The dataset contains over 500,000 retail transactions from multiple countries, with the majority originating from the United Kingdom.
2. Several products consistently recorded high sales volumes, indicating strong customer demand for home décor and gift-related items.
3. Revenue generation is heavily concentrated in a small number of countries, led by the United Kingdom.
4. Sales exhibit seasonal patterns, with significant increases during the final months of the year.
5. Customer spending is unevenly distributed, with a relatively small group of customers contributing a large portion of total revenue.

Top Insights

Insight 1: The United Kingdom Dominates Sales Revenue

The Revenue by Country visualization shows that the United Kingdom contributes the largest share of total revenue. This indicates that the business is highly dependent on its domestic market and may benefit from expanding sales efforts in other countries.

Insight 2: A Small Number of Products Drive Most Sales

The Top-Selling Products bar chart reveals that a limited number of products account for a significant proportion of total units sold. These products should be prioritized for inventory management and promotional campaigns.

Insight 3: Sales Increase During the Holiday Season

The Monthly Sales Trend line chart shows a noticeable rise in revenue during November and December. This suggests strong seasonal demand and highlights the importance of preparing inventory and marketing campaigns ahead of peak periods.

Insight 4: High-Value Customers Generate Significant Revenue

The Customer Spending visualization indicates that a small segment of customers contributes disproportionately to overall sales. Loyalty programs and personalized marketing strategies could help retain these valuable customers.

Insight 5: Product Popularity and Revenue Are Not Always the Same

Products with the highest sales quantities are not necessarily the highest revenue generators. Some lower-volume products generate substantial revenue because of their higher prices, emphasizing the importance of analyzing both sales volume and revenue.